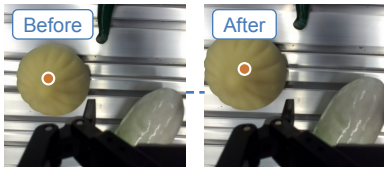
Body relations

motion response · acting part · configuration

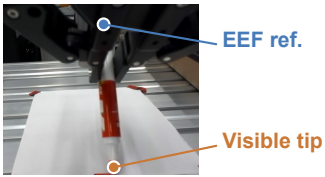
(a) Motion–image response



Recorded EEF move + image response

Use: choose direction (same camera)

(b) EEF reference and acting tip

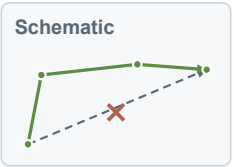


Condition: tool mounting / grasp

Use: locate the functional part

(c) Computed path check





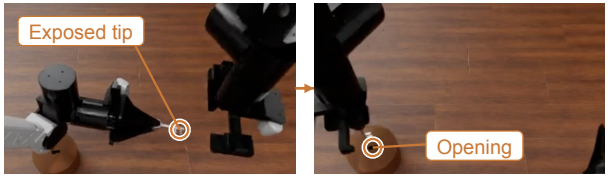
Declared geometry + current joints

Use: check the selected path

External interactions

grasp layout · downstream access · retention

(d) Grasp layout preserves downstream access



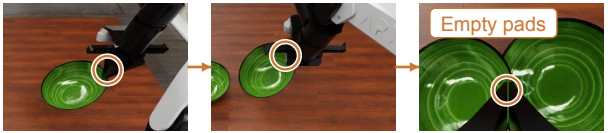
Grasp layout — Insertion access

Keep the working end accessible

Match grasp and target geometry

Demonstration candidate

(e) Retention across lift and transfer



Lift retained — Transfer retained — Later loss

Context: recorded bowl-rim grasp attempts
Observed: lift retention did not always persist
Use: recheck the bowl–pad relation at arrival
Recorded sim knowledge · cause unresolved